Machine Learning (BCS5-0705)

Module 1: Introduction: Machine Learning Basics, Hypothesis space and inductive bias, training and test set, Cross-Validation. Introduction to Statistical Learning: Bayesian Method.

Machine Learning: Supervised (Regression, Classification) vs. Unsupervised (clustering) Learning.

Data Preprocessing: Imputation, Outlier management, One hot encoding, Dimensionality Reduction- feature extraction, Principal Component Analysis (PCA), Singular Value Decomposition.

Supervised Learning: Regression- Linear regression, Polynomial Regression, Classification- Logistic Regression, K-nearest neighbor classifier.

[Books: ① Tom M. Mitchel, Machine Learning, TMH
② Alpaydin E. Introduction to machine learning. MIT press]

Introduction: Machine Learning Basics: Machine Learning is a branch of Artificial Intelligence (AI) where computer learns from data, without explicitly programming. It involves algorithms that analyze data, identify patterns and make predictions or decisions on new, unseen data. The core component of a machine learning model are representation, evaluation and optimization.

Key Concepts

① Data: Machine learning algorithms are trained on data, which can be structured (like spreadsheets) or unstructured (like images or text).

② Algorithms: These are the mathematical models that learns from the data. Examples includes linear regression, decision trees and neural networks.

③ Training: The process of feeding data to the algorithm to learn patterns and relationship.

④ Prediction/Decision: Once trained the algorithm can be used to make predictions or decisions on new, unseen data.

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5. Evaluation: Assessing the performance of the trained model using metrics like accuracy, precision and recall.

Types of Machine learning:

① Supervised Learning: The algorithm learns from the labeled data, where both input and desired output are provided (e.g. classifying emails as spam or not spam).

② Unsupervised learning: The algorithm learns from unlabeled data, discovering hidden patterns and structures (e.g. grouping patterns based on purchasing behavior).

③ Reinforcement learning: The algorithm learns through trial and error, receiving reward for correct actions and penalties for incorrect ones.

Hypothesis and Inductive bias

In machine learning, the hypothesis space is the set of all possible hypothesis (functions) that a learning algorithm can consider, given the training data and the algorithm's structure. The inductive bias is the set of assumptions that the learning algorithm used to guide its search within the hypothesis space and make predictions on unseen data. Essentially the inductive bias is the prior knowledge or tendency that the algorithm brings to the learning process, influencing which hypothesis it finds most plausible.

Elaboration

Hypothesis space: Imagine a learning algorithm trying to learn a relationship between inputs and outputs. The hypothesis space is the collection of all possible functions that the algorithm could learn. For eg. if the algorithm is learning to classify images as cats or dogs, the hypothesis space might include all possible functions that map images to either cat or dog.

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Inductive Bias: The inductive bias is what helps the algorithm choose one specific function from this vast space. It is the set of assumptions that the algorithm makes about which hypothesis are more likely to be correct. For instance an algorithm might have an inductive bias that favors simpler models (like linear models) over more complex ones (like high-degree polynomials).

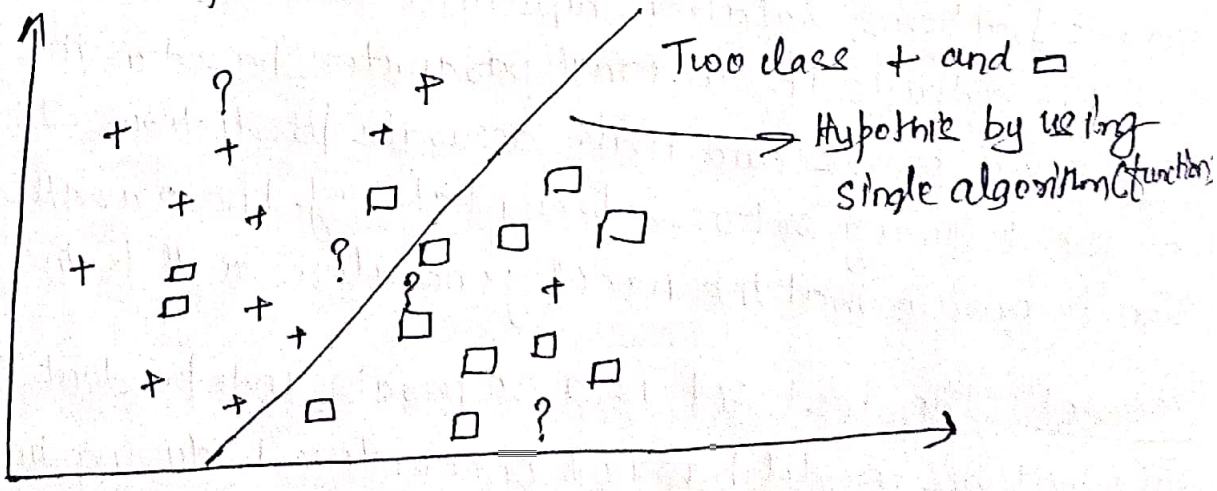
Why is inductive bias important?

① Generalization: Inductive bias helps the algorithm generalize well to unseen data by preventing it from overfitting to the training data.

② Efficiency: By narrowing down the search space, inductive bias makes the learning process more efficient.

③ Impact on performance: The choice of inductive bias significantly impacts the performance of the learning algorithm.

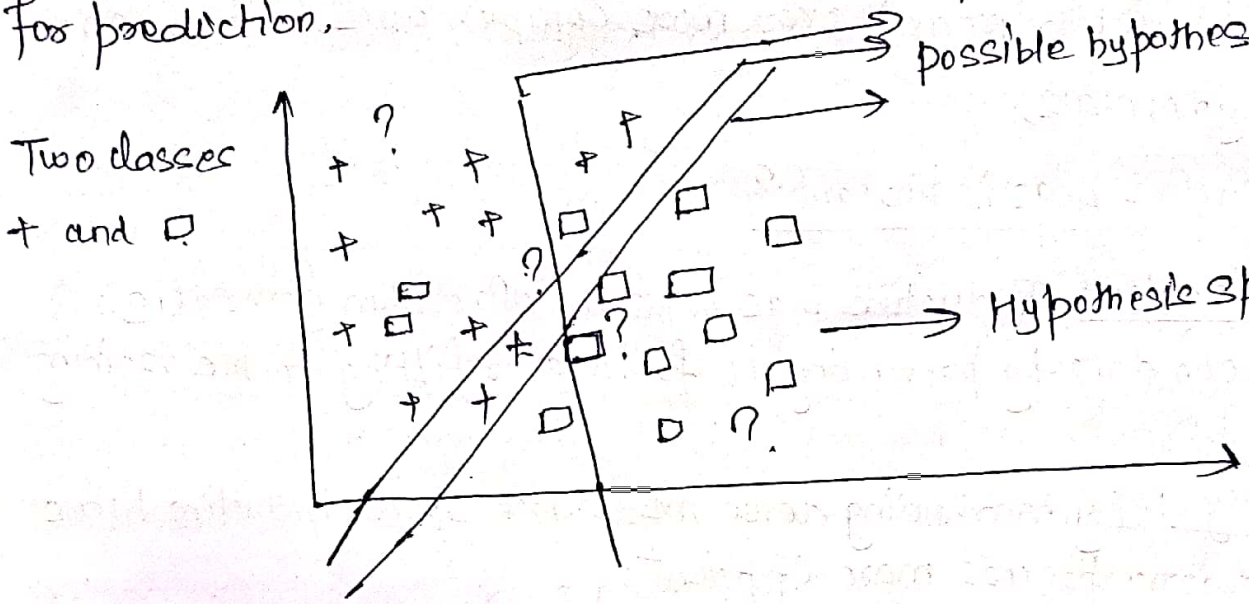
Considering an example: Let's say, we have a dataset with some attributes and some rows. We will plot all values in the graph and the graph will look as follows.



To predict the class of a new data point we will come up with a hypothesis function,

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We can also divide the data point class with some different hypothesis. We can come up with more than one hypothesis and the collection of all hypothesis is a hypothesis space. The following image shows the hypothesis space The machine learning model will choose the best (only one) hypothesis from the hypothesis space for prediction.



Training and Test sets: In machine learning training and test sets are fundamental divisions of a dataset used to develop and evaluate a model's performance.

Training Set: The training set is the portion of the data used to train the machine learning model. During this phase, the model learns patterns, relationships and features within the data. The algorithm adjusts its internal parameters based on the training data to minimize errors and make accurate predictions. It is crucial that the training set is representative of the overall data distribution to ensure that the model generalizes well to the unseen data.

Test Set: The test set is a separate, independent portion of the data that the model has not encountered during the training phase. It's purpose is to provide an unbiased evaluation of the trained model's performance on new, unseen data. By assessing the model's predictions on the test set, one can gauge its ability to generalize the.

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real world scenarios and identify issues such as overfitting, where the model performs well on training data but poorly on new data. The test set should ideally be used once after the model has been trained and its hyperparameters tuned.

Overfitting: Overfitting in machine learning occurs when a model learns the training data too well, including its noise and random fluctuations, leading to poor performance on unseen data.

Causes

① Too small training dataset: If the training data is limited, the model might learn specific quirks of that data instead of general trends.

② Noisy data: Irrelevant information in the training data can be learned by the model, confusing it.

③ Complex model: A model that is too complex for the data can easily overfit by memorizing the training data.

④ Long training time: Training for too long on the same data can lead to overfitting.

Cross Validation: Cross validation in machine learning is a resampling procedure used to evaluate the performance of a machine learning model and assess its ability to generalize the unseen data. It is a crucial technique for preventing overfitting and obtaining a more reliable estimate of a model's performance than a single train-test split.

working: The core idea of a cross validation involves partitioning the available dataset into multiple subsets or folds. The model is then iteratively trained and evaluated using different combinations of these folds.

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(a) Data partitioning: The original dataset is divided into a specified number of folds (eg, in K-fold cross-validation, it is divided into K equal-sized folds).

(b) Iterative Training and Testing: In each iteration one fold is designated as the validation (or test) set, and the remaining K-1 folds are used as the training set.

(c) Model evaluation: The model is trained on the training set and then evaluated on the unseen validation set. This process is repeated 'K' times, with each fold serving as the validation set exactly once.

(d) Performance aggregation: The performance metrics (eg. accuracy, precision, recall and F1-score) from each iteration are collected and averaged to provide a more robust and less biased estimate of the models overall performance.

Introduction to Statistical Learning

Bayes Method (Theorem): Bayes theorem is a fundamental concept in probability theory that plays a crucial role in machine learning algorithms, especially in the fields of Bayesian statistics and probabilistic modelling. It provides a way to update probabilities based on new evidence or information. In the context of machine learning, Baye's theorem is often used in Bayesian inference and probabilistic models

The theorem can be mathematically expressed as

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

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where

$P(A|B)$ is the posterior probability of event A given event B

$P(B|A)$ is the likelihood of event B given event A.

$P(A)$ is the prior probability of event A.

$P(B)$ is the total probability of event B.

Key terms related to Bayes Theorem

① Likelihood ($P(B|A)$): Represents the probability of observing the given evidence (features), given that the class is true.

② Prior probability ($P(A)$): Represents the probability of a particular class before considering any features. It is estimated from the training data.

③ Evidence Probability ($P(B)$): This is the probability of observing the given evidence (features).

④ Posterior Probability ($P(A|B)$): This is the updated probability of the class given the observed features.

Example

Determine if an email is spam based on the presence of a specific word "offer".

Given Probabilities

① $P(Spam)$: Prior probability of an email being spam = .30 (30% of emails are spam).

② $P(Non-Spam)$: Prior probability of an email being non-spam = .70 (70% of emails are non-spam).

③ $P(offer|Spam)$: Likelihood of "offer" appearing in a spam email = .80 (80% of spam emails contain offer).

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P(offer | Non-spam): Likelihood of "offer" appearing in a non-spam email = .10 (10% of non-spam email contain "offer").

Goal: Calculate P(Spam | offer), the posterior probability that an email is spam given that it contains the word "offer".

Formula = $P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$

A = Spam
B = offer

$= P(\text{Spam}|\text{offer}) = \frac{P(\text{offer}|\text{Spam}) \cdot P(\text{Spam})}{P(\text{offer})}$

P(offer) [Evidence]:

The probability of "offer" appearing in any email can be calculated using the law of total probability.

$P(\text{offer}) = P(\text{offer}|\text{Spam}) \cdot P(\text{Spam}) + P(\text{offer}|\text{Non-Spam}) \cdot P(\text{Non-Spam})$

$\Rightarrow (.80 \times .30) + (.10 \times .70) = .24 + .07 = .31$

$\therefore \boxed{P(\text{offer}) = .31}$

Finally Calculating P(Spam | offer) (Posterior):

$\therefore P(\text{Spam}|\text{offer}) = \frac{.80 \times .30}{.31} = \frac{.24}{.31} = .774$

If an email contains the word "offer", the probability that it is spam increases significantly from the prior probability of 30% to 77.4%. This demonstrates how Bayesian methods update Beliefs (prior probabilities) based on new evidence (observed data)

Machine Learning :

① **Supervised Machine Learning:** Supervised machine learning is a fundamental approach for machine learning and artificial intelligence. It involves training a model using labeled data, where each input comes with a corresponding correct output.

Working of Supervised Learning

The supervised learning algorithm consists of input features and corresponding output features. The process looks through:

ⓐ **Training data:** The model is provided with a training dataset that includes input data (features) and corresponding output data (lables or target variables).

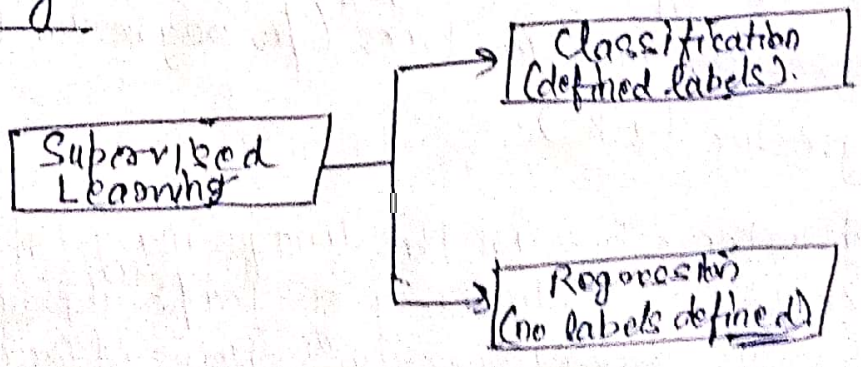
ⓑ **Learning process:** The algorithm processes the training data, learning the relationships between the input features and the output labels. This is achieved by adjusting the model's parameters to minimize the difference between its predictions and the actual labels.

After training, the model is evaluated using a test dataset to measure its accuracy and performance. Then the model's performance is optimized by adjusting parameters and using techniques like cross-validation to balance bias and variance. This ensures the model generalizes well to new, unseen data.

Types of Supervised Learning

ⓐ Classification

ⓑ Regression



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Classification: To predict a categorical or discrete output label for a given input. This means assigning data points to predefined categories or classes.

Examples

(a) Determining if an email is 'spam' or 'not spam'.

(b) Classifying an image as containing 'cat' or 'dog'.

(c) Predicting whether a customer will 'churn' or 'not churn'.

Common algorithms

Logistic Regression, Support Vector machines (SVM), Decision Trees, Random Forests, K-Nearest Neighbors (KNN), Naive Bayes.

Regression: To predict a continuous numerical value for a given input. This involves learning the relationship between input features and a continuous target variable.

Examples

(a) ~~To predict a continuous numerical value~~ Predicting house prices based on features like size, location and number of bedrooms.

(b) Forecasting stock prices or temperature.

(c) Estimate a person's age based on certain characteristics.

Common algorithms

Linear Regression, Polynomial Regression, Ridge Regression, Lasso Regression, Decision Trees (for regression tasks), Random Forests (for regression tasks).

While these two are the two primary types, supervised learning encompasses a wide array of specific algorithms designed to tackle these classification and regression problems efficiently.

② Unsupervised Machine Learning:

Unsupervised learning is a type of machine learning that analyzes unlabeled data to find hidden patterns, structures, and relationships without prior knowledge or guidance. Unlike supervised learning, which relies on labeled data, unsupervised learning algorithms discover inherent structures within the data by themselves. This often involves techniques like clustering and dimensionality reduction.

Key Concepts:

① **Unlabeled Data**: Unsupervised learning algorithms work with data that doesn't have predefined categories or labels.

② **Pattern discovery**: The core goal is to identify patterns, trends and relationships within the data without explicit instructions.

③ **No supervision**: Unlike supervised learning, there are no correct answers or target outputs for the algorithm to learn from.

Common Techniques of Unsupervised learning

① **Clustering**: Grouping similar data points together based on their characteristics.

② **Dimensionality Reduction**: Reducing the number of input variables while preserving important information.

③ **Anomaly Detection**: Identifying data points that deviate significantly from the norm.

④ **Association Rule Learning**: Discovering relationship between items in a dataset.

Applications

① Fraud Detection.

② Image and Video Analysis.

③ Customer Segmentation

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Data Preprocessing:

Data Imputation: Data is an essential component of any analysis, but it is not uncommon for datasets to contain missing values. Missing data can arise from various reasons, such as data entry errors, participant non-response or technical malfunctions. These missing values can lead to a biased or inefficient estimates, compromises the accuracy of statistical analysis. One way to address this issue is through data imputation techniques, which allow us to estimate missing values based on the available data. The goal of data imputation is to replace missing data with plausible values, based on the available information in the dataset. This process allows to retain as much information as possible from the incomplete data and provides with a more complete dataset for analysis.

Data Imputation Techniques

① **Mean Imputation Technique:** Mean imputation is a common and simple technique for filling in missing data. It involves replacing missing values with the mean value of non-missing values in the same feature/column. While this technique is easy to implement, it can lead to biased and imprecise estimates, if there is a high percentage of missing values.

② **Mode Imputation Technique:** Mode imputation is similar to mean computation, but instead of using the mean value, it uses the most frequent value in the same feature/column to impute the missing values. This technique is useful for imputing categorical data.

③ **Next or Previous Value:** One of the simpler methods for imputing missing values is to use the previous or next observed value.

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This technique is often referred to as 'forward fill' or 'backward fill' in pandas, where you either propogate the previous value forward ('forward fill') or use the subsequent value to fill the missing one ('backward fill').

④ Maximum or minimum value: Another approach to handling missing values is to replace them with the maximum or minimum value of the dataset or feature. This method might be helpful when trying to emphasize extreme values, especially in datasets where outliers or extreme values have significant meaning.

⑤ Hot Deck Imputation Technique: Hot deck imputation is a method that replaces missing values with a randomly selected value from a non-missing value in the same feature / column. The values in the dataset are grouped into categories based on similar attributes and each category forms a "deck". The missing values are then replaced with randomly selected values from the same deck.

⑥ Cold Deck Imputation Technique: Cold deck imputation is similar to hot deck imputation, but instead of randomly selecting values from the same deck, it selects values from different dataset or historical record.

⑦ Regression Imputation Technique: Regression imputation is a more advanced technique that involves using a regression model to predict missing values based on the non-missing values in the same dataset. This technique can lead to more accurate imputation, but it requires more computational resources and assumes that data follows a linear pattern.

⑧ K-Nearest Neighbor (KNN) Imputation Technique: KNN imputation involves finding the K nearest neighbor of each missing value based on

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the similarity of their non-missing values. The missing values are then imputed using the average of the K nearest neighbors. The technique can handle non-linear relationships between variables and useful for datasets with a small percentage of missing values.

Outlier Management

Outliers: Outliers are data points that are significantly different from the majority of data in a dataset. Think of them as students in a class who, for some reason, stand out. They might have exceptionally high or low scores compared to their classmates.

For eg, imagine you're analyzing the salaries of people in an office. Most employees earn between $30,000 to $60,000, but one person, let's say the CEO, has a salary of $10 million. That $10 million figure is an outlier because it is far removed from the rest of the data.

Why are outliers problematic?

Outliers can skew results, especially in machine learning, where they often mislead the model into finding inaccurate patterns. In simple terms, when you train a machine learning model with outliers, the model might "over-focus" on these outliers, leading to poor predictions.

For eg: Imagine building a model to predict housing prices. Most houses in your dataset are valued between $100,000 to $500,000, but there a few luxury properties worth millions. If your model includes these high-priced homes, it might predict higher than actual prices for regular houses.

Outliers can:

1. Distort the mean and standard deviation: They can make the average value much higher or lower than it should be.

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~~5/06/25~~ ~~Cluster-6D (Operating System)~~

Z-score example

The formula for Z-score is

$$z = \frac{x - \mu}{\sigma}$$

x — data point
μ — mean
σ — standard deviation

Calculating the mean and standard deviation of the datapoints

[25, 28, 30, 32, 35, 40, 42, 45, 120, 30, 29].

Mean(μ) = 38.73, standard deviation (σ) = 25.07

For the data point 25, the Z-score is

$$z = \frac{25 - 38.73}{25.07} = -0.55$$

$$\left[\text{Mean} = \frac{\sum x_i}{n}\right]$$

$$\left[\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}\right]$$

For the data point 120, the Z-score is

$$z = \frac{125 - 38.73}{25.07} = 3.24$$

The range for outlier is from [-3 to +3]

Therefore 25 is not an outlier but 120 is outlier.

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Inter Quartile Range (IQR) example

Data = [3, 4, 5, 2, 1, 7, 6, 10, 8, 9, 25]

Sorting in ascending order

Data = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 25]

Calculating the quartiles

Q_2 (Median) = 6

Q_1 (First Quartile) = [1, 2, 3, 4, 5] ∴ Median = 3

Q_3 (Third Quartile) = [7, 8, 9, 10, 25] ∴ median = 9

Interquartile Range (IQR)

$$IQR = Q_3 - Q_1 = 9 - 3 = 6.$$

Calculating the outlier fences:

Lower Fence : $Q_1 - (1.5 * IQR) = 3 - (1.5 * 6) =$

$3 - 9 = -6$

Upper Fence = $Q_3 + (1.5 * IQR) = 9 + (1.5 * 6) = 9 + 9$

$= 18$

[So Range is [−6 to 18].

The data points falls outside the above range would be qualified as outlier.

[So 25 is the outlier]

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② Influence Regression lines : In linear regression, outliers can shift the regression line, making predictions inaccurate.

Identifying Outliers

There are several way to spot outliers in a dataset :

(1) Visualization

(a) Use box plots and scatter plots. A box plot shows outliers as points outside the "whiskers" of the plot.

(b) Scatter plots can highlight data points that deviate from a typical pattern.

(2) Statistical Methods :

(a) Z-score : Measures how far a data point is from the mean in terms of standard deviations. If a Z-score is higher than a certain threshold (typically ±3), it is considered an outlier.

(b) Interquartile Range (IQR) : This method calculates the range between the 1st quartile (25th percentile) and 3rd quartile (75th percentile). Any data point outside the range of 1.5 times the IQR above the 3rd quartile or below the 1st quartile is considered an outlier.

Handling Outliers

(1) Remove the outliers : In cases where outliers are simply errors or irrelevant to your analysis, you can remove them. For instance if a dataset records age and someone's age is 300 years, it is safe to remove this point since it is unrealistic.

(2) Transform Data : Apply transformations like logarithmic or square root transformations to reduce the impact of outliers. This approach is helpful in datasets with values that are widely, such as income data.

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③ Use Robust Algorithms: Certain algorithms like decision trees and random forests are less sensitive to outliers. When working with data that naturally includes outliers, such as financial data, these algorithms can help reduce the impact of outliers on the model.

④ Use Clipping: Clipping involves setting an upper or lower limit on values. For eg. you could clip ages at 100 if it is unlikely anyone in the dataset is older.

Example of Outlier Handling: Suppose we have a dataset of ages with a majority between 20 and 60, but one entry lists an age of 150. Here's how it might be handled.

(1). Visualization: First, we see that this entry stands out in a scatter plot.

(2). Z-score Calculation: Calculate the Z-score of the age. Exceeds 3 it confirms the age is an outlier.

(3) Remove or Clip: Since 150 is an unrealistic age, we could remove this entry or clip it at 100.

One-hot encoding: One hot encoding is a technique used in machine learning to convert categorical data into a numerical format that machine learning models can understand and process. It creates new binary columns for each unique category in a feature, where a 1 indicates the presence of that category and a 0 indicates its absence.

A big part of the preprocessing is something encoding. This means representing each piece of data in a way that the computer can understand, hence the name encode, which literally means "convert to [computer] code". There are many different ways of encoding such as Label Encoding, or One hot Encoding. Label

encoding is intuitive and easy to understand,

Label encoding: Let us assume we are working with categorical data, like cats and dogs. Looking at the name of label encoding, you might be able to guess that it encodes labels, where label is just a category (i.e. cat or dog), and encode means just giving them a number to represent that category (1 for cat and 2 for dog). By giving each category a number, the computer now knows how to represent them, since the computer knows how to work with numbers. But there's problem that makes it often not work for categorical data. The problem is that with label encoding, the categories now have natural ordered relationships. The computer does this because it is programmed to treat higher numbers as higher numbers; it will naturally give the higher number higher weights.

for eg: Imagine if you had 3 categories of foods: apples, chicken and broccoli. Using label encoding, assign each of these a number to categorize them: apples=1, chicken=2, and broccoli=3. But if the model internally needs to calculate the average across categories, it might do... 1+3=4/2=2. This means according to the model, the average of apples and broccoli (chicken) together is chicken. Obviously that line of thinking by the model is going to lead to it getting correlations completely wrong, so there is a need to introduce one-hot encoding.

One hot encoding: Rather than labeling things as a number starting from 1 and then increasing for each category, we will go for more

of a binary style of categorizing. Consider the diagrams given below.

Label Encoding

Food Name	Categorical #	Calories
Apple	1	95
Chicken	2	231
Broccoli	3	50

One hot encoding

Apple	Chicken	Broccoli	Calories
1	0	0	95
0	1	0	231
0	0	1	50

What's the difference? Our category were formerly rows, but now they are columns. Our numerical variable, calories has however stayed the same. A 1 in a particular column will tell the Computer ~~that~~ the correct category for that row's data. In other words, an additional binary column has been created for each category.

Like many things in machine learning, we won't be using this in every situation; it's not outright better than label encoding. It just fixes a problem that encounter with label encoding when working with categorical data.

Dimensionality Reduction:

Feature Extraction: feature extraction is an important technique used in machine learning and data analysis to transform raw data into a set of features that are easier for the algorithms to work with. By reducing the complexity of the data, it keeps only the important parts and discards the unnecessary details. This allows machines to process data more efficiently and helps improve the accuracy of models. In many fields like image processing, natural language processing and signal processing, raw data often

comes with multiple characteristics. Many of these might be redundant or irrelevant. Feature extraction helps simplify this data, retaining only the most useful information for analysis.

Why is Feature Extraction Important? Feature extraction is important due to several reasons:

① **Reduced Computation Cost**: Raw data, especially from images or large datasets can be very complex. Feature extraction makes this data simpler, reducing the computational resources needed for processing.

② **Improved Model Performance**: By focusing on key features, machine learning models can work with more relevant information, leading to better performance and more accurate results.

③ **Better Insights**: Reducing the number of features helps algorithms concentrate on the most important data, eliminating noise and irrelevant information which can lead to deeper insights.

④ **Prevention of Overfitting**: Models with too many features may become too specific to the training data, making them perform poorly on new data. Feature extraction reduces the risk by simplifying the model.

Difference between Feature Selection and Feature Extraction.

Feature Selection	Feature Extraction.
① Selecting a subset of relevant features from the original set [Definition]	① Transforming the original features into a new set of features
② Reduce dimensionality ← [Purpose] →	② Transform data into a more manageable or informative representation.
③ Filtering, wrapper methods, embedded methods ← [Process] ⟶	③ Signal processing, statistical techniques, transformation algorithms
④ Subset of selected features [Output]	④ New set of transformed features
⑤ Lower cost ← [Computational Cost] →	⑤ May be higher, especially for complex

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⑥ Retains interpretability of original features. [Interpretability]	May lose interpretability depending on transformation.

Applications of Feature Extraction

① Image Processing and Computer Vision:

ⓐ Object Recognition: Extracting features from images to recognize objects or patterns within them.

ⓑ Facial Recognition: Identifying faces in images or videos by extracting facial features.

ⓒ Image classification: Using extracted features for categorizing images into different classes or groups.

② Natural Language Processing (NLP):

ⓐ Text classification: Extracting features from textual data to classify documents or texts into categories.

ⓑ Sentiment Analysis: Identifying sentiment or emotions expressed in text by extracting relevant features.

ⓒ Speech Recognition: Identifying relevant features from speech signals for recognizing spoken words or phrases.

③ Biomedical Engineering:

ⓐ Medical Image Analysis: Extracting features from medical images (like MRI or CT scans) to assist in diagnosis or medical research.

ⓑ Biological Signal Processing: Analyzing biological signals (such as EEG or ECG) by extracting relevant features for medical diagnosis or monitoring.

ⓒ Machine Condition Monitoring: Extracting features from sensor data to monitor the condition of machines and predict failures before they occur.

Baye's Theorem Questions:

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Q1 In a neighbourhood, 90% children were falling sick due to flu and 10% due to measels and no other disease. The probability of observing rashes for measles is 0.95 and for flue is 0.08. If a child develop rashes, find the child's probability of having flue.

Solution

Let F: Children with flu
M: Children with measles
R: Children showing the symptom of rash

$P(F) = 90\% = 0.9$
$P(M) = 10\% = 0.1$
$P(R|F) = 0.08$
$P(R|M) = 0.95$

$\therefore P(F|R) = \frac{P(R|F)P(F)}{[P(R|M)P(M) + P(R|F)P(F)]}$

$P(F)$ ↙ $P(R|M)P(M)$ ↖ $P(R|\text{not } F)P(\text{not } F)$

$\therefore P(F|R) = \frac{0.08 \times 0.9}{(0.95 \times 0.1) + (0.08 \times 0.9)} = \frac{.072}{.095 + .072} = \frac{.072}{0.167} = 0.43$

$\therefore P(F|R) = 0.43.$

Q2 It is observed that 50% of mails are spam. There is a software that filters spam mail before reaching the inbox. If accuracy for detecting a spam mail is 99% and chances of tagging a non-spam email as spam email is 5%. If a certain mail is tagged as a spam find the probability that it is not a spam mail.

Sol: Let E_1 = event of spam mail.
E_2 = event of non-spam mail
A = event of detecting a spam mail.

Now, $P(E_1) = 0.5$ and $P(E_2) = 0.5$
$P(A|E_1) = 0.99$ and $P(A|E_2) = 0.05$

$\therefore P(E_2|A) = \frac{P(A|E_2)P(E_2)}{P(A|E_1)P(E_1) + P(A|E_2)P(E_2)}$

$\Rightarrow \frac{0.05 \times 0.5}{(.99 \times .5) + (.05 \times 0.5)} = \frac{0.025}{0.520}$

$= 5/104 = 4.8\%$

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One Hot Encoding questions

Q1 Convert the given data to One-Hot Encoded Data

Original data

Team	Points
A	25
A	12
B	15
B	14
B	19
B	23
C	25
C	29

One-Hot Encoded Data

Team A	Team B	Team C	Points
1	0	0	25
1	0	0	12
0	1	0	15
0	1	0	14
0	1	0	19
0	1	0	23
0	0	1	25
0	0	1	29

Q2 Convert the given data to One-Hot Encoded Data

Original Data

Gender	Location
Male	South
Female	North
Male	West
Male	East

One-Hot Encoded Data

Gender_Male	Gender_Female	Loc_South	Loc_North	Loc_West	Loc_East
1	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	0	0	0	0	1

Q3 Convert the following given data to one-hot Encoded Data.

Original Data

Color
Red
Red
Yellow
Green
Yellow

→

One-Hot Encoded Data.

Red	Yellow Green	Blue
1	0	0
1	0	0
0	1	0
0	0	1
0	0	0

Q4 Convert the following given data to One-Hot Encoded Data.

original data

Labels
Sunny
Rain
Wind

→

Sunny	Rain	Wind
1	0	0
0	1	0
0	0	1

One-Hot Encoded Data

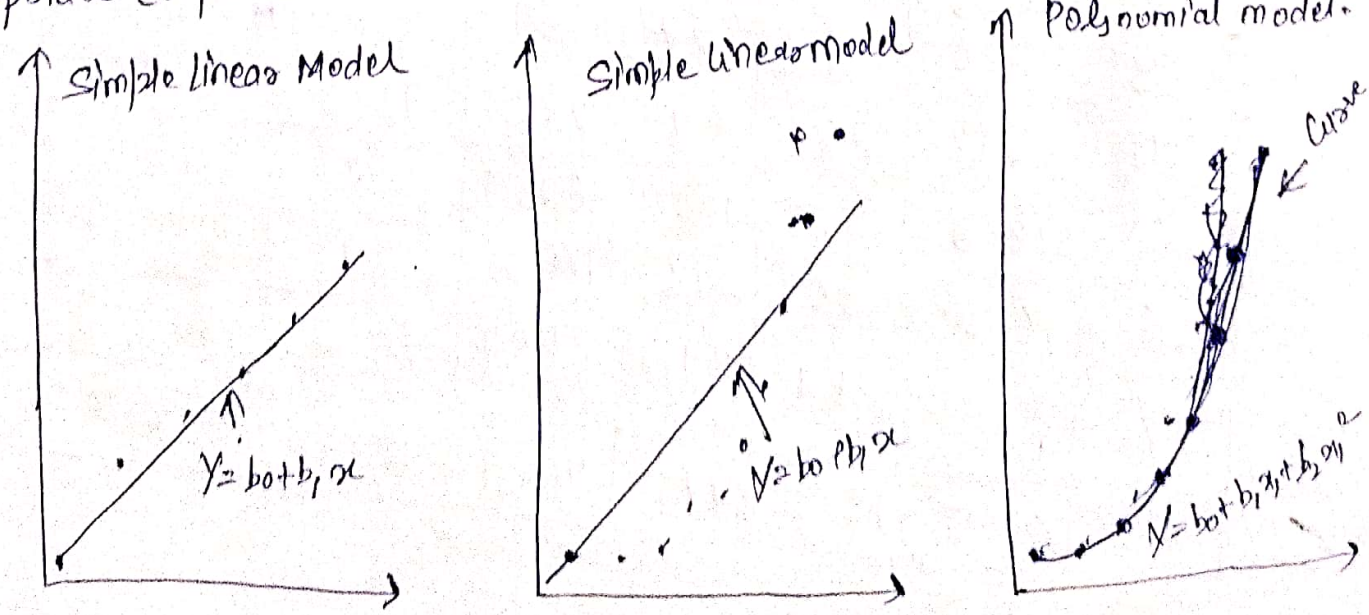
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Polynomial Regression: A simple linear regression algorithm only works when the relationship between the data is linear. But suppose we have non-linear data, then linear regression will not be able to draw a best-fit line. Simple regression analysis fails in such conditions. Polynomial regression is a form of regression analysis. In which the relationship between the independent variable x and the dependent variable y is modeled as an nth degree polynomial in x. It is also called the special case of Multiple Linear Regression in ML. Because we add some polynomial terms to the Multiple Linear Regression equation to convert it into Polynomial Regression.

In Polynomial regression, the original features are converted into Polynomial features of required degree $(2, 3, \dots, n)$ and then modeled using a linear model. The equation of Polynomial regression is as follows

$$y = b_0 + b_1 x_1 + b_2 x_1^2 + b_3 x_1^3 + \dots b_n x_1^n \quad \text{---(1)}$$

Polynomial regression offers greater flexibility compared to linear equation as it can model non-linearly separable data. However it requires some understanding of the data to select appropriate exponents that can lead to overfitting.

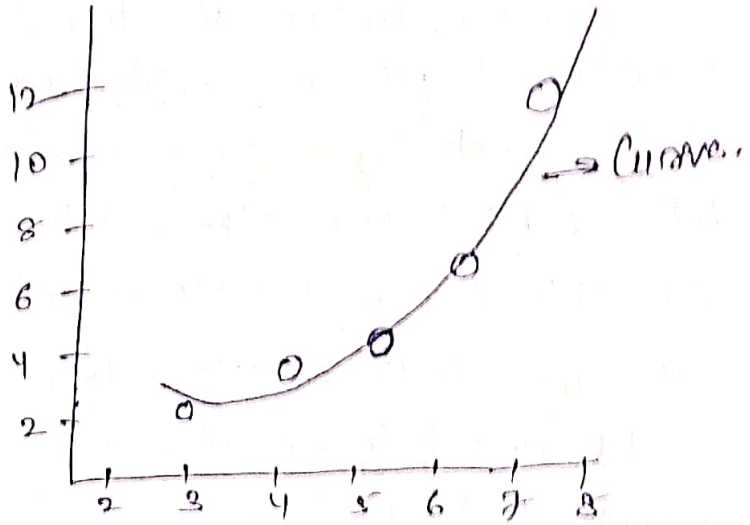


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Polynomial Regression Example

Consider the Data

X	Y
3	2.5
4	3.2
5	3.8
6	6.5
7	11.5



The quadratic polynomial regression model be

$$y = a_0 + a_1 x + a_2 x^2 \quad \text{---(2)}$$

The value of a_0, a_1 and a_2 are calculated using the following system of equations:

$$\sum y_i = n a_0 + a_1 \left(\sum x_i\right) + a_2 \left(\sum x_i^2\right)$$

$$\sum y_i x_i = a_0 \left(\sum x_i\right) + a_1 \left(\sum x_i^2\right) + a_2 \left(\sum x_i^3\right)$$

$$\sum y_i x_i^2 = a_0 \left(\sum x_i^2\right) + a_1 \left(\sum x_i^3\right) + a_2 \left(\sum x_i^4\right)$$

x	y	x^2	x^3	x^4	yx	yx^2
3	2.5	9	27	81	7.5	22.5
4	3.2	16	64	256	12.8	51.2
5	3.8	25	125	625	19	95
6	6.5	36	216	1296	39	234
7	12	49	343	2401	80.5	563.5
25	27.5	135	775	4659	158.8	966.2

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Putting the values in the equation

$$\therefore \quad 27.5 = 5a_0 + 25a_1 + 135a_2 \quad —$$
$$158.8 = 25a_0 + 135a_1 + 775a_2$$
$$966.2 = 135a_0 + 775a_1 + 4659a_2$$

by solving these equations for a_0, a_1 and a_2, the values are

$$a_0 = 12.4285$$
$$a_1 = -5.5128$$
$$a_2 = 0.76428$$

Putting these values in the eq. 2

$$y = 12.42857 - 5.5128x + 0.76428x^2$$

↓

Polynomial Regression Model trained

OR

Polynomial Regression equation.

Putting new different values of x [y can be predicted]

predicted → y or x

x = 8, y = 17.25

x = 9, y = 24.39

Covariance and Correlation of a given data

Covariance ⇒ Covariance quantifies how two variables X and Y vary with respect to their respective means. It is the average product of their deviations from the mean

Covariance

X	Y
10	40
12	48
14	56
8	32

$$Cov(X,Y) = \frac{\sum_{i=1}^{N} (x_i - \bar{x})(y_i - \bar{y})}{N}$$

$$\Rightarrow \frac{(-1)(-4) + (1)(4) + (3)(12) + (-3) \times (-12)}{4}$$

$$\Rightarrow \frac{4 + 4 + 36 + 36}{4} = \frac{80}{4} = 20$$

$\therefore$ **Cov(X, Y) = 20**

$\bar{X} = 11$ (mean)

$\bar{Y} = 44$ (mean)

① The positive covariance indicates as one variable increase, the other also increase.

② The Negative covariance signifies that as one variable increases, the other tends to decrease ③ Near-Zero covariance suggests that there is little or no linear relationship between two variables.

Correlation explained after Logistic Regression Example

Logistic Regression : Logistic Regression is a supervised learning algorithm used to estimate the probability that a given instance belongs to a particular class. It works by modeling the relationship between a categorical dependent variable and one or more independent variables. Logistic Regression is particularly useful for binary outcomes, such as classifying whether an email is spam or not.

Unlike linear regression, which predicts continuous numerical values, logistic regression outputs probabilities between 0 and 1. It achives this by passing the linear combination of input features though sigmoid function, ensuring that predictions remain within a valid probability range.

Real-life applications of logistic Regression in Machine Learning

① Spam Detection : Logistic regression classifies emails as spam or non-spam based on features like subject line and sender.

② Medical Diagnosis : It predicts the presence or absence of a disease based on patient data.

③ Loan Default Prediction : Determines whether a loan applicant is likely to default.

Logistic Function - Sigmoid Function : The sigmoid function is the heart of the logistic Regression. It takes the linear output of a model and transforms it into a probability value between 0 and 1.

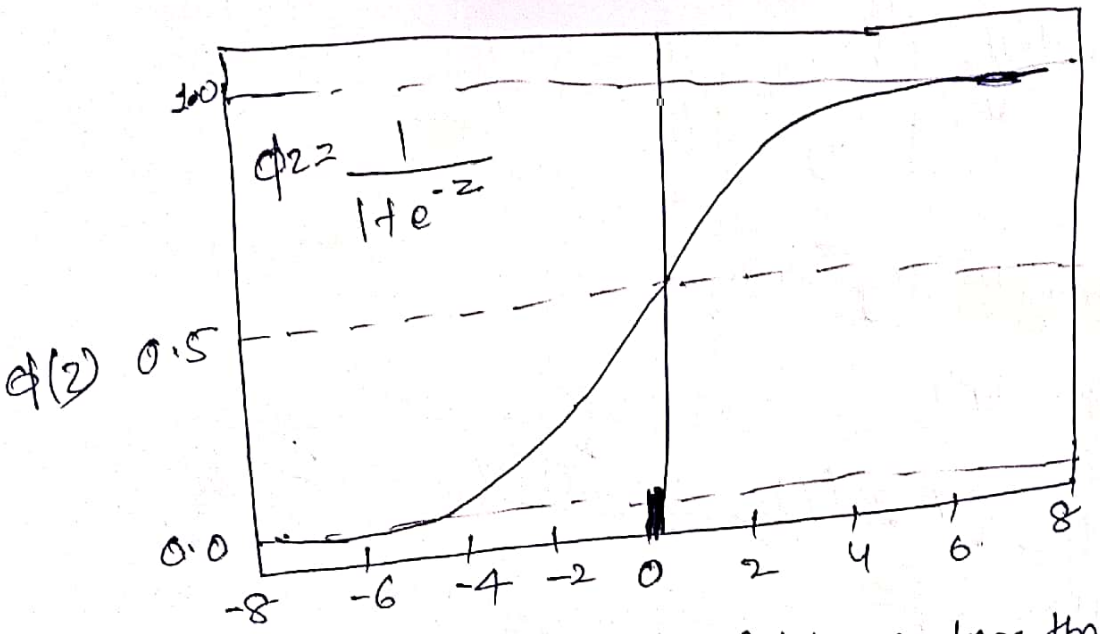
The sigmoid function is defined as

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

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where z is the linear combination of the input features.
This function maps any input value to a range between 0 and 1, which can be interpreted as the probability of the target variable belonging to a specific class.

Graphical Representation of the Sigmoid Curve



The sigmoid curve has an S-shape. At the center the output is 0.5 indicating equal probabilities for both classes. As z increases, the probability approaches 1 and as z decreases, the probability approches 0.

Types of Logistic Regression in Machine Learning

(A) Binomial Logistic Regression: Applied when the dependent variable has two outcomes (e.g, Spam vs non-Spam).

(B) Multinomial Logistic Regression: Used when the target variable has more than two categories without a natural order (eg, product categories).

(C) Ordinal Logistic Regression: Applied when the dependent variable has ordered categories (e.g, Satisfaction levels: low, medium, high)

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Logistic Regression Equation :

For Logistic regression, the input to the sigmoid function is the linear combination of the input features $x_1, x_2 \dots x_n$ and their respective weights.

$$z = m_1 x_1 + m_2 x_2 \dots m_n x_n + m_0$$

(intercept and the)

finally $h(z) = \frac{1}{1+e^{-z}}$ or $\sigma(z)$

Example of Logistic Regression

Consider the data

The following five persons have given some exams and their results.

Age (X)	Result (Y)
50	1
30	0
20	0
40	1
60	1

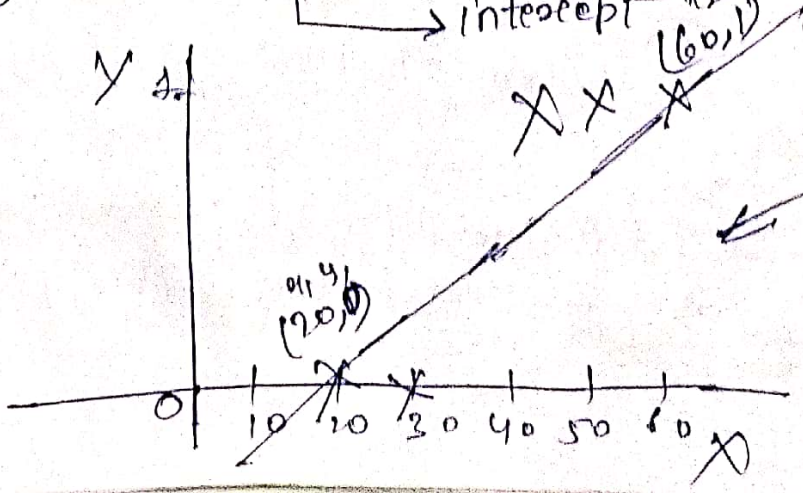
1 - pass
0 - fail.

$$h(z) = f(x) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-z}} \rightarrow (-b_0 + b_1 x_1)$$

$m_0 + m_1 x_1$ or $z = b_0 + b_1 x_1$ or $z = m_0 + m_1 x_1$

Solving for z first

$z = m_0 + m_1 x_1$ — slope; Independent feature (variable); dependent variable; intercept



X, Y points are (50,1) (30,0) (20,0) (40,1) (60,1).

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$y - y_1 = m(x - x_1)$ [Equation of line].

$$m = \frac{y - y_1}{x - x_1} = \frac{1 - 0}{60 - 20} = \frac{1}{40} = 0.025$$

$\therefore$ $y - y_1 = m(x - x_1)$

$y - 0 = 0.025(x - 20)$

$y = 0.025x - 0.5$

m_1 m_0.

[Mostly the value of m_0 and m_1 are given]

Now $\therefore$ $f(x) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(m_0 + m_1 x_1)}}$

$\Rightarrow$ $\frac{1}{1 + e^{-(-0.5 + (0.025)(x_1)}}$ → Here x_1 is the new data.

Suppose we have to predict for $x_1 = 15$ and 45 for pass(1) and fail(0).

$\therefore$ putting $x_1 = 15$

$$\therefore \frac{1}{1 + e^{-(-0.125)}} = \frac{1}{1 + e^{0.125}} = \frac{1}{1 + 1.1331}$$

$e = 2.718$

$\Rightarrow \frac{1}{2.1331} = 0.468$ ⟶ [for probability $P < 0.5$ Negative class, $P > 0.5$ Positive class].

Therefore, for age = 15, the result is 0 (fail)

putting $x_1 = 45$

$$\Rightarrow \frac{1}{1 + e^{-[-0.5 + (0.025)(45)}} = \frac{1}{1 + e^{-0.625}} = \frac{1}{1.5352} = 0.65$$

Therefore for age = 45, the result is 1 (pass).

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Correlation: is the statistical measurements that are used to quantify the strength of linear relationship between two variables. Correlation determines if two variables have a linear relationship.

The formula for Correlation is defined as

$$r(X,Y) = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n(\Sigma x^2) - (\Sigma x)^2][n(\Sigma y^2) - (\Sigma y)^2]}}$$

Considering an example

X	Y
1	2
2	4
3	4
4	6
5	7

[The Correlation values nearest to 1 suggest a strong Correlation between the variables]

Strong Correlation → 0.7 to 1.0 (or -0.7 to -1.0)
Moderate Correlation → 0.3 to 0.7 (or -0.3 to -0.7)
Weak Correlation → 0 to 0.3 (or -0 to -0.3)

X	Y	X^2	Y^2	XY
1	2	1	4	2
2	4	4	16	8
3	4	9	16	12
4	6	16	36	24
5	7	25	49	35
Σ 15	23	55	121	81

By putting values in the equation above

$$\Rightarrow \frac{(5\times 81) - (15)\times 23}{\sqrt{(5\times 55 - 225)(5\times 121 - 529)}} \Rightarrow \frac{60}{\sqrt{50\times 76}} = \frac{60}{\sqrt{3800}} = \frac{60}{61.64} = 0.9733$$

[0.9733 suggest a strong Correlation between X and Y]

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Question on Covariance and Correlation

Q: Find the Covariance and Correlation of the following Data:
$\{X = 6, 5, 3, 4, 2\}$ and $Y = \{12, 10, 8, 6, 4\}$

Sol: $\boxed{Cov(X, Y) = 3.2}$ and $\boxed{r(X, Y) = .9}$

Confusion Matrix Basics

① True Positive (TP): The model correctly predicted the positive class

② True Negative (TN): The model correctly predicted the negative class.

③ False Positive (FP): The model incorrectly predicted a positive class when it was negative.

④ False Negative (FN): The model incorrectly predicted a negative class when it was positive.

The formulas for different Performance metrics are defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1 Score} = \frac{2 * (\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})}$$

The Confusion Matrix is defined as:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

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Question: A model outputs 6TP, 6TN, 3FP and 2FN. Calculate the Recall. [Ans = .714]

Question: A model outputs 3TP, 4TN, 2FP and 1FN. Calculate the Precision. [Ans = 0.6]

Calculate the Accuracy, Precision, Recall and F1 score for the following Confusion Matrix

The Confusion matrix	Reality: 1	Reality: 0
Prediction : 1	75	5
Prediction : 0	5	15

Accuracy = 0.9, Precision = .9375, Recall = .9375, F1 Score = .9375

Question: Imagine a model trained to classify emails as 'Spam' [positive class] or 'Not Spam' [Negative class]. After testing the model on a dataset of 200 emails, the following results are obtained.

Actual spam emails: 80
- (a) Correctly identified as Spam (TP): 60
- (b) Incorrectly identified as Not Spam (FN): 20

Actual Not Spam emails: 120
- (a) Correctly Identified as Not Spam (TN): 100
- (b) Incorrectly identified as Spam (FP): 20

Answer the following Questions

(1) Construct the Confusion Matrix

(2) Compute the Accuracy, Precision, Recall and F1-Score.

Sol: (1) [Confusion matrix] →

	Predicted: Spam	Predicted: Not Spam
Actual: Spam	60 [TP]	20 [FN]
Actual: Not Spam	20 [FP]	100 [TN]

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Precision = $\frac{TP}{TP+FP} \Rightarrow \frac{60}{60+20} = \frac{60}{80} = \frac{3}{4} = .75 = 75\%$

Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} \Rightarrow \frac{60+100}{60+100+20+20} = \frac{160}{200} = .8 = 80\%$

Recall = $\frac{TP}{TP+FN} = \frac{60}{80} = \frac{3}{4} = .75 = 75\%$

F1 Score = $\frac{2 * (Precision * Recall)}{Precision + Recall} \Rightarrow \frac{2 * .75 \times .75}{.75 + .75} \Rightarrow \frac{1.125}{1.5}$

$\Rightarrow .75 = 75\%$

So Accuracy = 80% and Precision, Recall and F1-Score is 75%

Q on PCA (Principal Component Analysis): If a dataset comprises 2d points: (1, 0), (0, 1), (2, 2) and (4, 4). Compute the eigenvalues of its covariance matrix.

Sol: Calculating the mean of each dimension (Step 1)

The mean of x-coordinates and y-coordinates are calculated.

The x-coordinates are $x = \{1, 0, 2, 4\}$

The y-coordinates are $y = \{0, 1, 2, 4\}$

The mean of x is $\bar{x} = \frac{1+0+2+4}{4} = \frac{7}{4} = 1.75$

The mean of y is $\bar{y} = \frac{0+1+2+4}{4} = \frac{7}{4} = 1.75$

Step 2: Center the Data: The data points are centered by subtracting the respective means from each coordinate.

(1) The centered x-coordinates are

$x' = (x - \bar{x}) \Rightarrow \{1-1.75, 0-1.75, 2-1.75, 4-1.75\} = \{-0.75, -1.75, 0.25, 2.25\}$

(2) The centered y-coordinates are.

$x' = (y - \bar{y}) \Rightarrow \{0-1.75, 1-1.75, 2-1.75, 4-1.75\} = \{-1.75, -0.75, 0.25, 2.25\}$.

Step 3: Compute the covariance matrix

Covariance Matrix (C) = $\begin{bmatrix} C_{11} & C_{12} \\ \text{var}(x) & \text{cov}(x,y) \\ \text{cov}(x,y) & \text{var}(y) \\ C_{21} & C_{22} \end{bmatrix}$

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$C_{11} = \frac{\sum (x_i - \bar{x})^2}{n-1} \Rightarrow \frac{.5625 + 3.0625 + .0625 + 5.0625}{3} = \frac{8.75}{3}$

$= 2.9167$

$C_{22} = \frac{\sum (y_i - \bar{y})^2}{n-1} \Rightarrow \frac{(-1.75)^2 + 3.0625 + 0.5625 + .0625 + 5.0625}{3}$

$\Rightarrow \frac{8.75}{3} = 2.9167$

$C_{12} = C_{21} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1} \Rightarrow \frac{1.3125 + 1.3125 + .0625 + 5.0625}{3} = \frac{7.75}{3}$

$= 2.5833$

Now putting the Values in C

$\therefore C = \begin{bmatrix} 2.9167 & 2.5833 \\ 2.5833 & 2.9167 \end{bmatrix}$

$Var = \frac{\sum (x_i - \bar{x})^2}{n-1}$

$Cov = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$

Step 4 :-

Now Computing the Eigen Values for the C.

$\det(C - \lambda I) = 0$, where I is the identity matrix.

$$\det \begin{pmatrix} 2.9167 - \lambda & 2.5833 \\ 2.5833 & 2.9167 - \lambda \end{pmatrix} = 0$$

$\Rightarrow (2.9167 - \lambda)^2 - (2.5833)^2 = 0$

$(2.9167 - \lambda)^2 = (2.5833)^2$

$(2.9167 - \lambda = \pm 2.5833)$ [will be written as]

$\therefore \lambda_1 = 2.9167 + 2.5833 = 5.5$

$\lambda_2 = 2.9167 - 2.5833 = 0.3334$

$\therefore$ The Eigen Values are (5.5, 0.3334)

the measure of similarity or dissimilarity between data point

K-Nearest Neighbor Classifier (KNN): The KNN is a non-parametric supervised learning classifier, which uses proximity to make classifications or predictions about the grouping of an individual data point. It is one of the popular and simplest classification and regression classifiers used in machine learning today.

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While the KNN algorithm can be used for either regression or classification problems, it is typically used as classification algorithm, working off the assumption that similar points can be found near one another.

For classification problems, a class label is assigned on the basis of a majority vote - i.e. the model that is most frequently represented around a given data point is used. While this is technically considered "plurality voting", the term "majority vote" is more commonly used in literature. The distinction between these terminologies is that "majority voting" technically requires a majority of greater than 50%, which primarily works when there are only two categories. When you have multiple classes e.g four categories, you don't necessarily need 50% of the vote to make a conclusion about a class; you could assign a class label with a vote of greater than 25%.

Regression problem use a similar concept as classification problem, but in this case, the average of the K nearest neighbors is taken to make a prediction about a classification. The main distinction here is that classification is used for discrete values, where Regression is used for continuous ones. However before a classification can be made, the distance must be defined. Euclidean distance is most commonly used, which we'll delve into more below.

It is worth noting that the KNN algorithm is also part of a family of "lazy learning" models, meaning that it only stores a training dataset versus undergoing a training stage. This means that all the computation occurs when a classification or prediction is being made.

The Distance metrics used to Compute KNN : In order to determine which data points are closest to a given query point, the distance between the query point and the other data points will need to be calculated. These distance metrics help to form decision boundaries, which partitions query points into different regions.

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The some of the distance metrics used for computing the distance are

① **Euclidean distance**: This is the most commonly used distance measure and is limited to Real-valued vectors. The formula is defined below.

$$\text{Euclidean distance} = d(x, y) = \sqrt{\sum_{i=1}^{n} (y_i - x_i)^2} \quad \text{---(1)}$$

x_i and y_i calculate the distance between x_i and y_i [the two coordinates]

② **Manhattan distance**: This is another popular distance metric, which measures the absolute value between two points. It is also referred to as taxicab distance or city block distance. The formula is defined below

$$\text{Manhattan distance} = d(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i| \right) \quad \text{---(2)}$$

③ **Minkowski distance**: This distance measure is general form of Euclidean and Manhattan distance metrics. The parameter, p, in the below formula, allows for the creation of other distance metrics. Euclidean distance is represented by this formula when p is equal to 2 and Manhattan distance when p is equal to one.

~~Minkowski distance = $d(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i| \right)^{1/p}$~~

$$\text{Minkowski distance} = d(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i| \right)^{1/p} \quad \text{---(3)}$$

The equation 1 for two data coordinates is written as.

$x_i = (\underset{x_1}{2}, \underset{y_1}{4})$, $y_i = (\underset{x_2}{3}, \underset{y_2}{4})$

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(3-2)^2 + (4-4)^2}$$

$$= \sqrt{(1)^2 + (0)^2} = \sqrt{1} = 1.$$

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Example of KNN

(Q) Predict if a new fruit is apple based on "Roundness score" and "Sweetness score". Use K=3 and Euclidean distance.

Sol: Dataset (Training Data):

Fruit Type	Roundness score (X_1)	Sweetness score (X_2)
Apple	8	9
Apple	7	8
Orange	5	7
Orange	6	6
Apple	9	7

New Data point to classify: A fruit with a roundness score of 7.5 and sweetness score of 7.5. (K=3 means, considering the 3 closest neighbors)

① Calculating distance between new data point and each fruit features.

∴ New data point = (7.5, 7.5),
Apple(1) = (8, 9); Apple(2) = (7, 8), Orange(1) = (5, 7),
Orange(2) = (6, 6) and Apple(3) = (9, 7).

Using the Euclidean distance formula, between

(7.5)(7.5) (x_1 y_1) and Apple (8, 9) (x_2 y_2) $\Rightarrow \sqrt{(x_2-x_1)^2+(y_2-y_1)^2} = \sqrt{(8-7.5)^2+(9-7.5)^2}$
$= \sqrt{0.25+2.25} = \sqrt{2.5} = 1.58$

(7.5)(7.5) and Apple (7, 8) $\Rightarrow \sqrt{(7-7.5)^2+(8-7.5)^2} = \sqrt{0.25+0.25} = \sqrt{0.5} \Rightarrow 0.71$

(7.5)(7.5) and Orange (5, 7) $\Rightarrow \sqrt{(5-7.5)^2+(7-7.5)^2} = \sqrt{6.25+0.25} = \sqrt{6.5} = 2.55$

(7.5)(7.5) and Orange (6, 6) $\Rightarrow \sqrt{(6-7.5)^2+(6-7.5)^2} = \sqrt{2.25+2.25} = \sqrt{4.5} = 2.12$

(7.5)(7.5) and Apple (9, 7) $\Rightarrow \sqrt{(9-7.5)^2+(7-7.5)^2} = \sqrt{2.25+0.25} = \sqrt{2.5} = 1.58$

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② Identify K-Nearest Neighbors: The K=3 and for K=3 the 3 nearest neighbors to new data point are

Apple(7, 8) – Distance = 0.71
Apple(8, 9) – Distance = 1.58
Apple(9, 7) – Distance = 1.58

③ Classify: Determine the majority class among the K-nearest neighbors. In this example, all three nearest neighbors are Apple. Therefore the new data point (fruit) is classified as Apple.

Residuals and Mean Square Error (MSE) in Linear Regression

Residuals: In linear regression, residuals are the differences between the the actual and the predicted values for each data point, representing the error of a specific prediction.

$$\therefore \text{Residuals} = y - \hat{y}$$

$\therefore$ y is the actual value and $\hat{y}$ is the value predicted by the regression model for a given data point.

Residuals measure how far each data point deviates from the regression line, serving as an estimate of the unobservable statistical error.

Mean Square Error (MSE): MSE is the average of the squared residuals across all data points in the dataset.

$$MSE = \frac{\sum (y_i - \hat{y}_i)^2}{n}$$

Low MSE value indicates a more accurate model with less error. MSE can range from 0 to positive infinity, with a value of 0 representing a perfect fit where all predictions match the actual values.

The units of MSE are the squared units of the dependent variable (the y-axis).

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Example

Imagine you have a simple linear regression model that predicts a person's exam score based on the number of hours they studied. The model's equation is:

$$\text{Predicted score } (\hat{y}) = 10 \times \text{Hours studied } (x) + 20$$

10 → slope; 20 → intercept

Hours (studied) (x)	Actual score (y)	Predicted score ($\hat{y}$)	Residuals ($y - \hat{y}$)
1	28	$10(1)+20 = 30$	$28-30 = -2$
2	45	$10(2)+20 = 40$	$45-40 = 5$
3	52	$10(3)+20 = 50$	$52-50 = 2$
4	58	$10(4)+20 = 60$	$58-60 = -2$

Negative Residual: For the first and fourth student, the actual score was 2 points lower than the model predicted. Means the model overestimated the predicted point.

Positive Residual: For the second student, the actual score was five times higher than the predicted one. Means the model underestimated the predicted point. For the third student, the actual score was two times higher than the predicted one. Means the model underestimated the predicted point.

Magnitude: Magnitude identifies the difference between the actual data points and predicted values. Small residuals magnitudes (close to 0) indicates a better fit. Large magnitudes may suggest issues like inaccurate predictions or presence of outliers that need further investigation or model adjustements.

$$MSE = \frac{\sum (y_i - \hat{y}_i)^2}{m}$$

MSE for the above Example

$$\Rightarrow \frac{(-2)^2 + (5)^2 + (2)^2 + (-2)^2}{4} \Rightarrow \frac{4+25+4+4}{4} \Rightarrow \frac{37}{4}$$

$$\Rightarrow \boxed{9.25}.$$

The Mean Square Error should be as minimum as possible for perfect prediction.

Cross Validation methods in machine learning

Common Cross-Validation Techniques

① **Hold Out Method** : The simplest method, where the dataset is split into a training set and test set (e.g., 70% of training and 30% of testing). The model is trained on the training set and evaluated on the test set.

② **K-Fold Cross Validation** : The dataset is divided into k equally sized "folds". In each iteration one fold is used as the test set, and the remaining k-1 folds are used as the training set. The process is repeated k times, ensuring that every data point serves as the part of the test set exactly once. The performance metrics from each fold are averaged to provide a more robust estimate of the model's performance. Provides a more reliable and stable estimate of the model performance compared to the holdout method.

③ **Leave-One-Out Cross-Validation (LOOCV)** : A special case of K-Fold where k equals the number of data points (n). In each iteration, a single data point is used as the test set, and the remaining n-1 datapoints are used for training. Maximizes the amount of data used for training in each fold. Computationally expensive for large datasets due to n iterations.

④ **Stratified K-Fold Cross-Validation** : Similar to K-Fold, but ensures that each fold maintains the same proportion of target variable classes as the original dataset. Particularly useful for imbalanced datasets.

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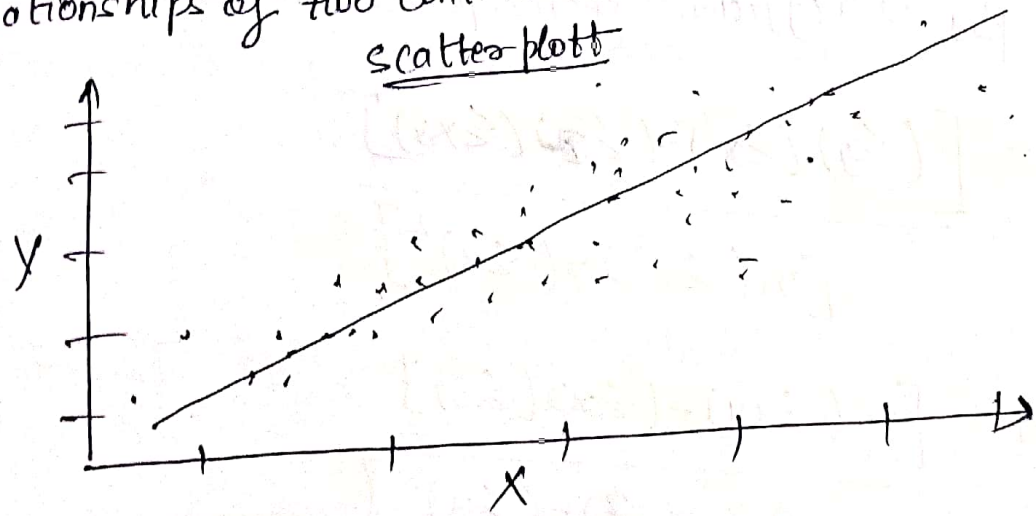
BCSE0133: Machine Learning Lab: Linear Regression

(1). Estimate parameters of a model based on Linear Regression method using a given set of training data.

Solut (1). Simple linear Regression is used to find out the best relationship between a single input variable (independent variable) and output variable (dependent variable) provided that both variables are continuous in nature. This relationship represents how an input variable is related to the output variable and how it is represented by a straight line.

To understand this concept, let us have a look at scatter plots. Scatter diagram or plots provides a graphical representation of the relationships of two continuous variables.

scatter plott



After looking at scatter plot we can understand:

① Direction ② The Strenght ③ Linearity.

The above characteristics are between variable Y and variable X. The above scatter plot shows us that variable X and X possesses a strong positive linear relationship. Hence we can project a straight line which can define the data in the most accurate way possible. If the relationship between variable X and variable Y is strong and linear, then we can conclude that particular independent variable X is effective input variable to predict dependent variable Y.

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Steps to implement Simple Linear Regression.

1. Analyze data (analyze scatter plot for linearity)
2. Get sample data for model building
3. Then design a data that explains the data.
4. Use the same developed method on whole population to make predictions.

The equation that represents how an independent variable X is related to a dependent variable Y.

$$Y = \beta_0 + \beta_1 X$$

$\beta_0 \Rightarrow$ Intercept, calculated over dependent variable Y

$\beta_1 \Rightarrow$ coefficient of independent variable X, also called slope.

$$\beta_0 = \frac{[(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

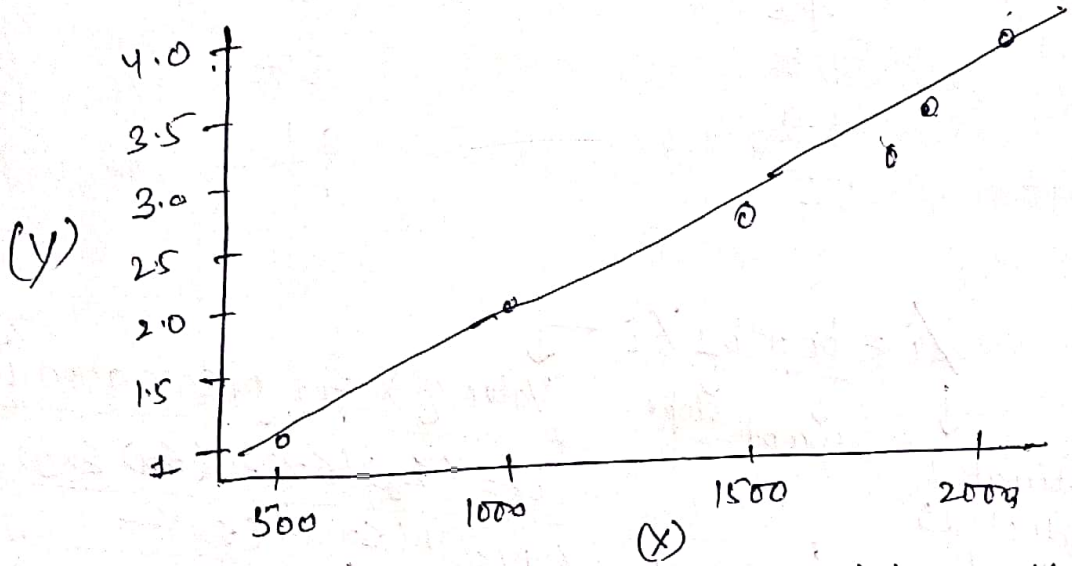
$$\beta_1 = \frac{[n(\Sigma xy) - (\Sigma x)(\Sigma y)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

Considering an example. We want to predict the weight gain based upon calories Consumed only based on the below given data.

Calories Consumed (X)	Weight Gain (Y)
500	1
1000	2
1500	2.5
1750	3.2
2100	3.9

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Now, if we want to predict weight gain when you consume 2500 calories. First we need to visualize the data by drawing a scatter plot of the data to conclude that Calories Consumed is the best independent variable X to predict dependent variable Y.



Now substituting the x value (2500) in equation of regression model as:

$$y = \beta_0 + \beta_1 * x$$
$$= \beta_0 + \beta_1 * 2500$$

The intercept (β_0) and (β_1) are computed from the previous formulas

· So $y = \beta_0 + \beta_1 x$ → Calculated from the formulas

and $\beta_0 = 0.1233623$ and $\beta_1 = .0017494$

$\therefore y = 0.1233623 + .0017494 * 2500$

$\therefore y = 4.4968623$

So weight gain predicted by our simple linear regression is 4.49 kgs after consumption of 2500 calories.

Example 2: (Linear Regression)

This tutorial will understand how to use a regression equation to predict the glucose level, given the age.

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Subject	Age (X)	Glucose (Y)
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81
7	55	?

Note

$\hat{Y}_i = b_0 + b_1 X_i$

- $\hat{Y}_i$: Estimated (or predicted) Y value for observation i
- b_0: intercept
- b_1: slope
- X_i: Value of X for observation i

$$b_0 = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$b_1 = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

Subject	Age(X)	Glucose (Y)	XY	x^2	y^2
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

$$\therefore b_0 = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$= \frac{(486 \times 11409) - (247 \times 20485)}{(6 \times 11409) - (247)^2} = \frac{1848974 - 65914}{7445}$$

$$\therefore \boxed{b_0 = 65.14}$$

Now $b_1 = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2} = \frac{(6 \times 20485) - (247 \times 486)}{(6 \times 11409) - (247)^2} = \frac{2868}{7445}$

$$= 0.385$$

$$\therefore \boxed{b_1 = 0.385}$$

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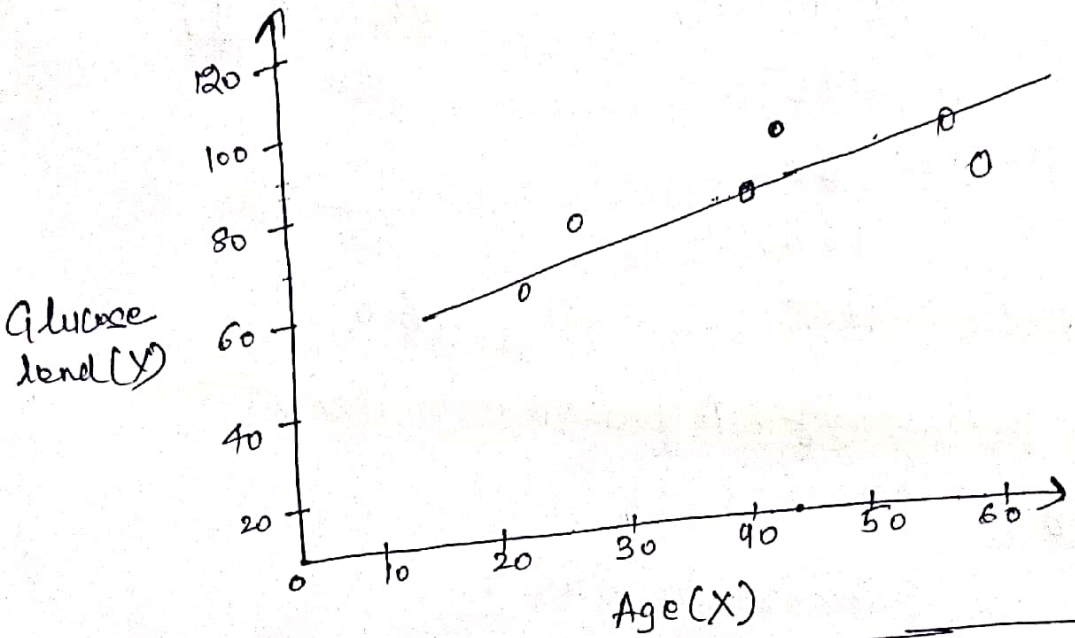
Now inserting the values in the equation.

$y = 65.14 + (0.385335 * x) = 65.14 + (0.385225 * 55)$.

$y = 86.327$

∴ The prediction value of y for a given ($x = 55$) is 86.327

The graph of above example is as follows



(2). Estimate parameters of a model based on maximum likelihood estimation method using a given set of training dataset.

Sol: In the last page

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Singular Value Decomposition (SVD):

The Singular Value Decomposition of a matrix is a factorization of that matrix into three matrices. It has some interesting algebraic properties and conveys important geometrical and theoretical insights about linear transformations. It also has some important applications in data science.

Mathematics behind SVD:

The SVD of mxn matrix A is given by the formula $A = U \Sigma V^T$

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where

① U: m×m matrix of the orthonormal (orthogonal and normalized) eigenvectors of AA^T.

② V^T: transpose of a n×n matrix containing the orthonormal eigenvectors of A^TA.

③ Σ: diagonal matrix with r elements equal to the root of the positive eigen values of AA^T or A^TA (both matrices have the same positive eigen values anyway).

Example of a SVD

(Q) Find the Singular Value decomposition of the matrix $A=\begin{bmatrix}4 & 0\\ 3 & -5\end{bmatrix}$, the 2×2 matrix.

Sol: The formula is ~~A=UΣV^T~~ ∴ $A=U\Sigma V^T$

∴ V^T = Transpose of a n×n matrix containing the orthonormal eigenvectors of A^TA

$$\therefore A=\begin{bmatrix}4 & 0\\ 3 & -5\end{bmatrix} = A^TA=\begin{bmatrix}4 & 3\\ 0 & -5\end{bmatrix}\begin{bmatrix}4 & 0\\ 3 & -5\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}25 & -15\\ -15 & 25\end{bmatrix}$$

Now first, we have to calculate the eigen values of A^TA.

$$A^TA=\begin{bmatrix}25 & -15\\ -15 & 25\end{bmatrix}$$

$$A^TA-\lambda I=\begin{bmatrix}25-\lambda & -15\\ -15 & 25-\lambda\end{bmatrix} \quad \text{——— (1)}$$

orthonormal and normalized: each vector is perpendicular to each other and has unit length

$$\therefore \det(A^TA-\lambda I)=((25-\lambda)(25-\lambda))-(-15)\times(-15))$$

$$\Rightarrow 625-25\lambda-25\lambda+\lambda^2-225$$

$$\Rightarrow \lambda^2-50\lambda+400$$

$$\therefore \lambda^2-50\lambda+400=0$$

$$\therefore \lambda^2-40\lambda-10\lambda+400=0$$

$$\therefore \lambda(\lambda-40)-10(\lambda-40)=0$$

$$\therefore (\lambda-10)(\lambda-40)$$

∴ Two Eigen Values are 10, 40

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Once we have calculated the eigen values, it's time to calculate the two eigenvectors for each eigen value.

For eigen value = 10, the eigen vector calculation is as follows

Putting the $\lambda = 10$, in eqn 1

$$\therefore \quad A^TA - 10I = \begin{bmatrix} 25-10 & -15 \\ -15 & 20-10 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 15 & -15 \\ -15 & 15 \end{bmatrix}$$

In order to find the eigen vector, we need to find the nullspace of a matrix whose $AB = 0$. In other words

$$\begin{bmatrix} 15 & -15 \\ -15 & 15 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\therefore \quad R_2 \leftarrow R_2 + R_1$$

$$\Rightarrow \begin{bmatrix} 15 & -15 \\ 0 & 0 \end{bmatrix}$$

$$\text{Row } 1 \times 1/15 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore \quad v_1 - v_2 = 0$$

$$\boxed{v_1 = v_2} \quad \therefore \text{ for } v_1 = 1, \; v_2 \text{ is also } 1$$

$$\therefore \quad \text{The Eigen Vectors for } \lambda = 10 \text{ are } \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Now, Next step is to convert it into a unit vector. The way we do this is by taking the Columnar values and dividing them by taking the Square root of sum of squares of the values. So it can be expressed as

$$\sqrt{(1)^2 + (1)^2} = \sqrt{2}$$

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So final eigen vectors for $\lambda=10$ are

$$\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

Same steps are repeated for getting the eigen vector of a eigen value 40.

$$\therefore A^TA - 40I = \begin{bmatrix} 25-40 & -15 \\ -15 & 85-40 \end{bmatrix} = \begin{bmatrix} -15 & -15 \\ -15 & -15 \end{bmatrix}$$

$$\therefore R_2 \to R_2 - R_1 \Rightarrow \begin{bmatrix} -15 & -15 \\ 0 & 0 \end{bmatrix}$$

$$\therefore R_1 \to -1/15\, R_1 \Rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$v_1 + v_2 = 0$$

$$v_1 = -v_2 \quad \therefore \text{for } v_2 = 1 \text{ the } v_1 = -1$$

Therefore the Eigen vectors for Eigen Value 40 =

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

and Final eigen vectors after calculating the unit vector are

$$= \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

Now that we have got both the eigen vectors, let put it together

$$\begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = V$$

$$\begin{bmatrix} \sqrt{40} & 0 \\ 0 & \sqrt{10} \end{bmatrix} = \Sigma$$

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Note that the diagonal values in sigma are always in the descending order and so the vectors are also placed in that corresponding order.

Now that we have our V and Sigma matrix, now it's time to find. U. We can just multiply the equation by sigma (inverse) and V on both sides to get the equation for U. In this case, as V is an orthogonal matrix, the transpose and inverse of V are the same therefore V(transpose) multiplied by V becomes an Identity matrix.

$$\therefore AV\Sigma^T = U$$

$$AV = \begin{bmatrix} 4 & 0 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -4/\sqrt{2} & 4/\sqrt{2} \\ -8/\sqrt{2} & -2/\sqrt{2} \end{bmatrix}$$

Next, we convert it to unit vectors by using the same steps as defined earlier

$$AV = \begin{bmatrix} \frac{-4}{\sqrt{2}} & 4/\sqrt{2} \\ -8/\sqrt{2} & -2/\sqrt{2} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{-4}{\sqrt{2}} \times \frac{1}{2\sqrt{10}} & \frac{4}{\sqrt{2}} \times \frac{1}{\sqrt{10}} \\ \frac{-8}{\sqrt{2}} \times \frac{1}{2\sqrt{10}} & \frac{-2}{\sqrt{2}} \times \frac{1}{\sqrt{10}} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{-2}{2\sqrt{5}} & 4/2\sqrt{5} \\ \frac{-4}{2\sqrt{5}} & -2/2\sqrt{5} \end{bmatrix} = \begin{bmatrix} -1/\sqrt{5} & \frac{2}{\sqrt{5}} \\ -2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix}$$

Next we multiply this matrix with sigma (transpose) which is sigma in itself because its a diagonal matrix.

$$AV\Sigma^T = \begin{bmatrix} -1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix} \begin{bmatrix} \sqrt{40} & 0 \\ 0 & \sqrt{10} \end{bmatrix}$$

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$$= \begin{bmatrix} -\frac{1}{\sqrt{5}} \times \sqrt{10} & \frac{2}{\sqrt{5}} \times \sqrt{10} \\ -\frac{2}{\sqrt{5}} \times \sqrt{40} & -\frac{1}{\sqrt{5}} \times \sqrt{10} \end{bmatrix} = \begin{bmatrix} -\sqrt{\frac{10}{5}} & 2\sqrt{\frac{10}{5}} \\ -2\sqrt{\frac{40}{5}} & -\sqrt{\frac{10}{5}} \end{bmatrix} = \begin{bmatrix} -2\sqrt{2} & 2\sqrt{2} \\ -4\sqrt{2} & -\sqrt{2} \end{bmatrix}$$

Next, we need to convert this to unit vectors to get the final U matrix.

$$U = \begin{bmatrix} -2\sqrt{2} & 2\sqrt{2} \\ -4\sqrt{2} & -\sqrt{2} \end{bmatrix} = \begin{bmatrix} \frac{-2\sqrt{2}}{\sqrt{40}} & \frac{2\sqrt{2}}{\sqrt{10}} \\ \frac{-4\sqrt{2}}{\sqrt{40}} & \frac{-\sqrt{2}}{\sqrt{10}} \end{bmatrix} = \begin{bmatrix} -2\sqrt{\frac{2}{40}} & 2\sqrt{\frac{2}{10}} \\ -4\sqrt{\frac{2}{40}} & -\sqrt{\frac{2}{10}} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2/\sqrt{20} & 2/\sqrt{5} \\ -4/\sqrt{20} & -2/\sqrt{5} \end{bmatrix} = \begin{bmatrix} -1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix} \therefore U = \begin{bmatrix} -1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix}$$

So therefore we have calculated the U, Σ and V. and decomposed the matrix A into three matrices as given below.

$$A = U \Sigma V^T$$

$$\Rightarrow \begin{bmatrix} 4 & 0 \\ 3 & -5 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix} \begin{bmatrix} \sqrt{40} & 0 \\ 0 & \sqrt{10} \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Important properties of SVD: Some important properties of SVD. includes

1. **Uniqueness**: U, Σ and V are unique
2. **Orthonormality**: U and V are orthonormal matrices
3. **Diagonal matrix**: Σ is a diagonal matrix with positive, non zero diagonal entries that are sorted in descending order.
4. **Generalization**: SVD is a generalization of diagonalization to non-square matrices.
5. **Data Compression**: SVD can be used for data compression by retaining only the most significant singular values and their corresponding vectors.

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⑥ Image processing: SVD is an algebraic transform that can be used for image processing applications.

⑦ Scalability: SVD can process large datasets efficiently.

Principal Component Analysis (PCA): PCA is a statistical procedure that uses an orthogonal transformation to converts a set of correlated variables to a set of uncorrelated variables. PCA is the most widely used tool in exploratory data analysis and in machine learning for predictive models.

PCA is an unsupervised learning algorithm technique used to examine the interrelations among a set of variables. It is also known as general factor analysis where regression determines a line of best fit. The main goal of PCA is to reduce the dimensionality of the dataset while preserving the most important patterns or relationships between the variables without any prior knowledge of the target variables. PCA is used to reduce the dimensionality of the dataset by finding a new set of variables, smaller than the original set of variables, retaining most of the sample's information and useful for the regression and classification of data.

Steps of PCA for Image classification

① All gray scale images are transformed to column vectors. Then mean value is generated from these column vectors. The mean generation formula is illustrated below (Eq 1). C_i denotes the number of column vectors and M stores the mean value.

$$M = \frac{1}{n} \sum_{i=1}^{n} C_i \quad \text{---------- (1)}$$

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(2) The difference is generated between the column vectors and the mean value. Precisely the mean values are differentiated from the column vectors and stored in variable D as shown in eq. 2.

$$D = C_i - M \quad \text{---------- (2)}$$

(3) In third step, eigen values and eigen vectors are generated of $D \cdot D^T$. But $D \cdot D^T$ consumes memory in large amount so these are computed of $D^T \cdot D$. Then eigenface space is produced by considering the n topmost eigenvectors of the corresponding m topmost eigen values. Suppose these E largest eigenvectors are stored in variable L. In terms of eq, it is defined as:

$$L = [E_1 \dots E_n] \quad \text{---------- (3)}$$

(4) The eigenfaces are generated by using D and L. The variable F is used for storing the eigenfaces values. The eq is defined as

$$\therefore F = D \cdot L \quad \text{---------- (4)}$$

(5) Ultimately the images are projected in the feature space as shown in eq. 5. N represents the feature vectors of the projected images.

$$N = F^T \cdot D \quad \text{---------- (5)}$$

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